

MATH 2790 Fall 2025

Homework 1

Due date: Wednesday, Aug 27

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment is from sections 1.2–1.6.

Problems:

- 1. Question (1) on page 71 of textbook under section 1.3.
- 2. Question (3) on page 71 of textbook under section 1.3.
- 3. Question (4) on page 72 of textbook under section 1.4.
- 4. Question (1) on page 72 of textbook under section 1.5.
- 5. Question (3) on page 72 of textbook under section 1.5.
- 6. Question (6) on page 73 of textbook under section 1.5.
- 7. John deposits \$400 into an account earning interest based on annual discount rate of 4.7%. Find the balance at the end of seven years.
- 8. A savings account earns compound interest at an annual effective interest rate of i . Given that $i_{[2,5]} = 20\%$,
 - (a) Find i .
 - (b) Find the equivalent annual discount rate d .
 - (c) Find $d_{[1,4]}$.